26 Supersymmetric Field Theories

Appendix Majorana Spinors

This appendix summarizes some algebraic properties of Majorana spinors that are needed in dealing with superfields.

Consider a four-component fermionic Majorana spinor s that like Q or θ may be expressed in the form

$$s = \begin{pmatrix} e\zeta^* \\ \zeta \end{pmatrix}, \tag{26.A.1}$$

where ζ is some two-component spinor and e is the $2 \cdot 2$ matrix

$$e \equiv \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2 \,.$$

Such a spinor is related to its complex conjugate by

$$s^* = \begin{pmatrix} 0 & e \\ -e & 0 \end{pmatrix} s = -\beta\gamma_5\epsilon s \,, \tag{26.A.2}$$

where ϵ is the $4 \cdot 4$ matrix

$$\epsilon \equiv \begin{pmatrix} e & 0 \\ 0 & e \end{pmatrix} \tag{26.A.3}$$

and as usual γ_5 and β are $4 \cdot 4$ matrices

$$\gamma_5 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

with $\mathbf{1}$ and 0 understood here as $2 \cdot 2$ submatrices. Taking the transpose of Eq. (26.A.2) and multiplying on the right by β gives the equivalent formula

$$\bar{s} \equiv s^\dagger \beta = s^T \epsilon\gamma_5 \,. \tag{26.A.4}$$

The anticommutation of the spinor components limits the variety of covariants that can be formed from a Majorana spinor. To see this, it will be convenient first to consider the symmetry properties of bilinear covariants, which will be of some interest in themselves. For a pair of Majorana spinors s_1 and s_2 and any $4 \cdot 4$ numerical matrix M, Eq. (26.A.4) gives

$$\begin{aligned} \bar{s}_1 M s_2 &= \sum_{\alpha\beta} s_{1\alpha} s_{2\beta} (\epsilon\gamma_5 M)_{\alpha\beta} = -\sum_{\alpha\beta} s_{2\alpha} s_{1\beta} (\epsilon\gamma_5 M)_{\beta\alpha} \\ &= +\sum_{\alpha\beta} s_{2\alpha} s_{1\beta} (M^T \epsilon\gamma_5)_{\alpha\beta} = \bar{s}_2 (\epsilon\gamma_5)^{-1} M^T \epsilon\gamma_5 s_1 \,, \end{aligned}$$

with the minus sign following the second equal sign arising from the fermionic nature of these spinors. In Section 5.4 we found that the 16 covariant matrices formed from Dirac matrices satisfy

$$M^T = \begin{cases} +\mathcal{C}M\mathcal{C}^{-1}, & M = \mathbf{1},\ \gamma_5\gamma_\mu,\ \gamma_5, \\ -\mathcal{C}M\mathcal{C}^{-1}, & M = \gamma_\mu,\ [\gamma_\mu, \gamma_\nu], \end{cases} \tag{26.A.5}$$

where $\mathcal{C}$ is the matrix

$$\mathcal{C} = \gamma_2 \beta = -\epsilon\gamma_5 = \begin{pmatrix} -e & 0 \\ 0 & e \end{pmatrix} . \tag{26.A.6}$$

It follows that

$$(\bar{s}_1 M s_2) = \begin{cases} +(\bar{s}_2 M s_1), & M = \mathbf{1},\ \gamma_5\gamma_\mu,\ \gamma_5, \\ -(\bar{s}_2 M s_1), & M = \gamma_\mu,\ [\gamma_\mu, \gamma_\nu]. \end{cases} \tag{26.A.7}$$

In particular, setting $s_1 = s_2 = s$, we find that

$$\bar{s}\gamma_\mu s = \bar{s}[\gamma_\mu, \gamma_\nu] s = 0 \,, \tag{26.A.8}$$

so the only bilinear covariants formed from a single Majorana spinor s are $\bar{s}s$, $\bar{s}\gamma_5\gamma_\mu s$, and $\bar{s}\gamma_5 s$.

In considering the form of the most general superfields, we need to have expressions for the products of two or more Majorana spinors. For two spinors, we recall that any $4 \cdot 4$ matrix may be expanded as a sum of the 16 covariant matrices $\mathbf{1}$, γ_μ, $[\gamma_\mu, \gamma_\nu]$, $\gamma_5\gamma_\mu$, γ_5. Lorentz invariance tells us that for the matrix $s_\alpha \bar{s}_\beta$ this expansion must take the form

$$\begin{aligned} s\bar{s} = {} & k_S(\bar{s}s) + k_V \gamma_\mu (\bar{s}\gamma^\mu s) + k_T [\gamma_\mu, \gamma_\nu](\bar{s}[\gamma^\mu, \gamma^\nu] s) \\ & + k_A \gamma_5 \gamma_\mu (\bar{s}\gamma_5\gamma^\mu s) + k_P \gamma_5 (\bar{s}\gamma_5 s) \,, \end{aligned}$$

where the k's are constants to be determined. Eq. (26.A.8) shows that we may take $k_V = k_T = 0$. The remaining coefficients may be calculated by multiplying on the right with $\mathbf{1}$, $\gamma_5\gamma^\mu$, and γ_5 and taking the trace, which gives $k_S = -1/4$, $k_A = +1/4$, and $k_P = -1/4$. In this way we find that

$$s\bar{s} = -\frac{1}{4}(\bar{s}s) + \frac{1}{4}\gamma_5\gamma_\mu(\bar{s}\gamma_5\gamma^\mu s) - \frac{1}{4}\gamma_5(\bar{s}\gamma_5 s) \,. \tag{26.A.9}$$

By multiplying on the right with $-\epsilon\gamma_5$ and using Eq. (26.A.4), we may put this in the form

$$s_\alpha s_\beta = \frac{1}{4}(\epsilon\gamma_5)_{\alpha\beta}(\bar{s}s) + \frac{1}{4}(\gamma_\mu\epsilon)_{\alpha\beta}(\bar{s}\gamma_5\gamma^\mu s) + \frac{1}{4}\epsilon_{\alpha\beta}(\bar{s}\gamma_5 s) \,, \tag{26.A.10}$$

or, equivalently,

$$s_\alpha s_\beta = \frac{1}{4}(\epsilon\gamma_5)_{\alpha\beta}(s^T\epsilon\gamma_5 s) + \frac{1}{4}(\gamma_\mu\epsilon)_{\alpha\beta}(s^T\epsilon\gamma^\mu s) + \frac{1}{4}\epsilon_{\alpha\beta}(s^T\epsilon s) \,. \tag{26.A.11}$$

Now consider the product $s_\alpha s_\beta s_\gamma$ of three components of a Majorana spinor s. We can divide s into left- and right-handed parts

$$s = s_L + s_R \,, \qquad s_L = \frac{1}{2}(\mathbf{1} + \gamma_5)s \,, \qquad s_R = \frac{1}{2}(\mathbf{1} - \gamma_5)s \,. \tag{26.A.12}$$

Each of s_L and s_R has only two independent components, so since the square of any fermionic c-number vanishes, we have $s_{L\alpha}s_{L\beta}s_{L\gamma} = 0$ and $s_{R\alpha}s_{R\beta}s_{R\gamma} = 0$ for all α, β, and γ, and therefore

$$s_\alpha s_\beta s_\gamma = s_{L\alpha}s_{L\beta}s_{R\gamma} + s_{L\alpha}s_{R\beta}s_{L\gamma} + s_{R\alpha}s_{L\beta}s_{L\gamma} + L \leftrightarrow R \,,$$

with '$L \leftrightarrow R$' denoting the sum of previous terms with labels L and R interchanged. To evaluate this expression, we multiply Eq. (26.A.11) by suitable factors of $(\mathbf{1}+\gamma_5)/2$, and find

$$s_{L\alpha}s_{L\beta} = \frac{1}{4}[\epsilon(\mathbf{1}+\gamma_5)]_{\alpha\beta}(s_L^T\epsilon s_L)\,.$$

If we now multiply this with $s_{R\gamma}$, since $(s_R^T\epsilon s_R)s_{R\gamma} = 0$ we can drop the label L on the spinors in the bilinear $(s_L^T\epsilon s_L)$:

$$s_{L\alpha}s_{L\beta}s_{R\gamma} = \frac{1}{4}[\epsilon(\mathbf{1}+\gamma_5)]_{\alpha\beta}(s^T\epsilon s)s_{R\gamma}\,.$$

The same arguments also yield

$$s_{R\alpha}s_{R\beta}s_{L\gamma} = \frac{1}{4}[\epsilon(\mathbf{1}-\gamma_5)]_{\alpha\beta}(s^T\epsilon s)s_{L\gamma}\,.$$

Adding the sum of these two expressions to the same quantity with γ replaced with α or β yields finally

$$\begin{aligned} s_\alpha s_\beta s_\gamma = \frac{1}{4}(s^T\epsilon s)\big[&\epsilon_{\alpha\beta}s_\gamma - (\epsilon\gamma_5)_{\alpha\beta}(\gamma_5 s)_\gamma - \epsilon_{\alpha\gamma}s_\beta \\ &+ (\epsilon\gamma_5)_{\alpha\gamma}(\gamma_5 s)_\beta + \epsilon_{\beta\gamma}s_\alpha - (\epsilon\gamma_5)_{\beta\gamma}(\gamma_5 s)_\alpha\big]. \end{aligned} \tag{26.A.13}$$

To calculate the products of four Majorana spinor components, we note that $(s^T\epsilon s)$ contains only terms with two s_Ls or two s_Rs, so

$$(s^T\epsilon s)s_\gamma s_\delta = (s^T\epsilon s)[s_{R\gamma}s_{R\delta} + s_{L\gamma}s_{L\delta}]\,.$$

Using Eq. (26.A.11) to evaluate the sum in square brackets, and noting that

$$(s^T\epsilon s)(s^T\epsilon\gamma_5 s) = (s_L^T\epsilon s_L)(s_R^T\epsilon s_R) - (s_R^T\epsilon s_R)(s_L^T\epsilon s_L) = 0\,,$$

we find that

$$(s^T\epsilon s)s_\gamma s_\delta = \frac{1}{4}\epsilon_{\gamma\delta}(s^T\epsilon s)^2\,. \tag{26.A.14}$$

Multiplying Eq. (26.A.13) with s_δ therefore yields the result

$$\begin{aligned} s_\alpha s_\beta s_\gamma s_\delta = \frac{1}{16}(s^T\epsilon s)^2\big[&\epsilon_{\alpha\beta}\epsilon_{\gamma\delta} - (\epsilon\gamma_5)_{\alpha\beta}(\epsilon\gamma_5)_{\gamma\delta} - \epsilon_{\alpha\gamma}\epsilon_{\beta\delta} \\ &+ (\epsilon\gamma_5)_{\alpha\gamma}(\epsilon\gamma_5)_{\beta\delta} + \epsilon_{\beta\gamma}\epsilon_{\alpha\delta} - (\epsilon\gamma_5)_{\beta\gamma}(\epsilon\gamma_5)_{\alpha\delta}\big]. \end{aligned} \tag{26.A.15}$$

Any product of five components of s vanishes, so this completes the list of formulas for products of components of a Majorana spinor.

We can use these formulas to derive some additional relations that will be useful in dealing with superfields. By contracting Eq. (26.A.13) with $(\epsilon\gamma_5)_{\beta\gamma}$ and $(\epsilon\gamma_\mu)_{\beta\gamma}$, we find

$$s_\alpha(\bar{s}s) = -(\gamma_5 s)_\alpha(\bar{s}\gamma_5 s) \tag{26.A.16}$$

and

$$s_\alpha(\bar{s}\gamma_5\gamma_\mu s) = -(\gamma_\mu s)_\alpha(\bar{s}\gamma_5 s)\,. \tag{26.A.17}$$

From Eqs. (26.A.16) and (26.A.17) we may derive 'Fierz' identities

$$(\bar{s}s)^2 = -(\bar{s}\gamma_5 s)^2\,, \qquad (\bar{s}\gamma_5\gamma_\mu s)(\bar{s}\gamma_5\gamma_\nu s) = -\eta_{\mu\nu}(\bar{s}\gamma_5 s)^2\,. \tag{26.A.18}$$

Also, Eq. (26.A.14) may be put in a covariant form

$$(\bar{s}\gamma_5 s)s\bar{s} = -\frac{1}{4}\gamma_5(\bar{s}\gamma_5 s)^2\,. \tag{26.A.19}$$

It will also be useful to record the reality properties of bilinear products of Majorana spinors. For any pair of Majorana spinors s_1, s_2 satisfying the phase convention (26.A.1), Eqs. (26.A.2) and (26.A.4) give

$$(\bar{s}_1 M s_2)^* = -(s_1^\dagger \epsilon\gamma_5 M^* s_2^*) = (\bar{s}_1 \beta\epsilon\gamma_5 M^* \beta\epsilon\gamma_5 s_2)\,.$$

(The minus sign in the middle expression arises from reversing the interchange of s_1 and s_2 that occurs when we take the complex conjugate.) But Eqs. (5.4.40) and (26.A.6) give $\beta\epsilon\gamma_5\gamma_\mu^*\beta\epsilon\gamma_5 = \gamma_\mu$, so

$$\beta\epsilon\gamma_5 M^*\beta\epsilon\gamma_5 = \begin{cases} +M, & M = \mathbf{1},\ \gamma_\mu,\ [\gamma_\mu, \gamma_\mu], \\ -M, & M = \gamma_\mu\gamma_5,\ \gamma_5, \end{cases} \tag{26.A.20}$$

and therefore

$$(\bar{s}_1 M s_2)^* = \begin{cases} +(\bar{s}_1 M s_2), & M = \mathbf{1},\ \gamma_\mu,\ [\gamma_\mu, \gamma_\mu], \\ -(\bar{s}_1 M s_2), & M = \gamma_\mu\gamma_5,\ \gamma_5. \end{cases} \tag{26.A.21}$$

Finally, we mention that any spinor u may be written in terms of a pair of Majorana spinors $s_\pm$ as

$$u = s_+ + i s_-\,, \tag{26.A.22}$$

where

$$s_+ \equiv \frac{1}{2}(u - \beta\epsilon\gamma_5 u^*)\,, \qquad s_- \equiv \frac{1}{2i}(u + \beta\epsilon\gamma_5 u^*)\,. \tag{26.A.23}$$

To check that $s_\pm$ are Majorana spinors satisfying Eq. (26.A.2), it is only necessary to recall that $\beta\epsilon\gamma_5$ is real, and that $(\beta\epsilon\gamma_5)^2 = \mathbf{1}$.